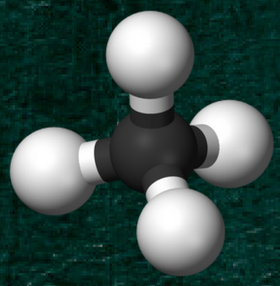
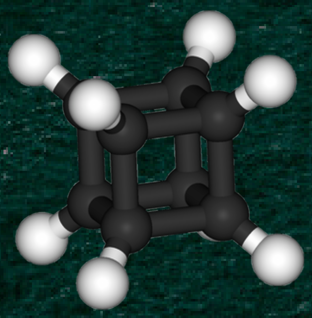
THINK OUTSIDE THE CUBANE



Methane

Geometry: Tetrahedral

Ideal bond angles: 109.5°



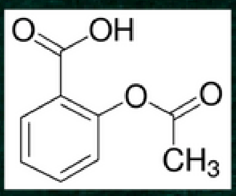
Cubane

Geometry: Tetrahedral

Bond angle: 90°

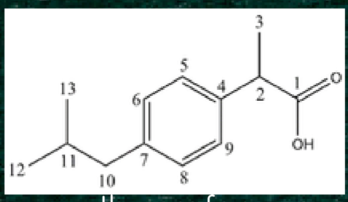
The hydrocarbon, cubane (C_8H_8), is unique due to its bond angle strain. First synthesised in 1964 from cyclopentan-2-one in a 13-step synthesis, Cubane is stable despite the extreme strain as it lacks decomposition routes, lending it high versatility!

How can Cubane improve current medicinals?



Paracetamol

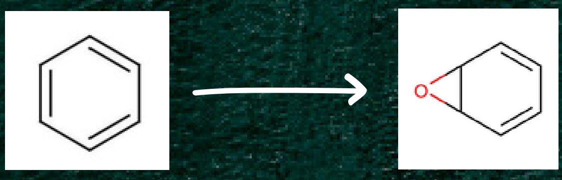
Aspirin



Ibuprofen

Aromaticity is common in medicines due to the added stability and existing chemistry surrounding aromatic compounds.

HOWEVER



Our body has enzymes that can oxidise benzene rings to benzene oxides, which are very reactive, causing toxic effects.

Aromatic molecules also do not degrade easily and can be environmentally toxic.

MEDICINAL PROPERTIES OF CUBANE

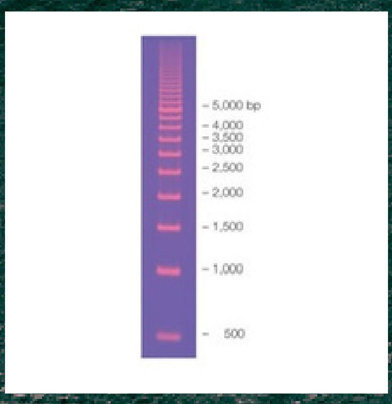
- High metabolic stability → rigid, strained structure
- Longer lasting effect of the drug in the body
- Enhanced solubility → enables easier drug absorption into the body
- 3D orientation improves how drugs fit into protein targets by attaching groups into complementary 3D positions

CUBANE'S GREAT POINTS

- Cubane is less toxic than benzene → Better for overdosing
- Longer half-life
- Higher binding selectivity and potency → Same amount, cubane is more effective
- Strong IP protection → Less likely to get sued

INSANE CUBANE

- Molecular ruler - Studying distance dependent interactions
- Molecular bearings in rotor stator system (fullerene cocrystals)
- Cancer and HIV treatments
- Born to be 109.5° forced to be 90°



CUBANE'S SHARP EDGES

- Compared to benzene, Cubane is very expensive to manufacture
- Benzene's chemistry is also very well researched and organic synthesis surrounding benzene is relatively simple
- Difficult to scale safely due to unpredictable reactions

In summary, scientific curiosity and challenging the impossible are crucial to progress. Although Cubane is not yet ready for widespread application, with persistence, it will undoubtedly become a game changing compound.

"The reasonable man adapts himself to the world; the unreasonable one persists in trying to adapt the world to himself. Therefore, all progress depends on the unreasonable man" - George Bernard Shaw